Agentic Theory III: Stimulus Tensor Propagation, Emergent High-Order Addictions, and the Tensorial Output Stream in Agentic Neural Networks

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Abstract

The two preceding papers in this series introduced the Artificial Junky Neuron (AJN) as an atomic agentic unit and established its composition into multi-layer Agentic Neural Networks (ANN-Ψ). The present work addresses two phenomena that emerge specifically from the network topology and cannot be observed at the single-unit level.

First, we formalize the *transformation and generalization of stimulus tensors across layers*: a low-level sensory tensor received by the first layer undergoes successive recodings—abstraction, fusion, and re-projection—as it propagates upward through an ANN-Ψ. Each AJN layer constructs a new, higher-dimensional stimulus tensor from the praxic outputs of the layer below, creating a stimulus hierarchy. We show that this hierarchy supports the emergence of *high-order addictions*: stable craving attractors at intermediate and deep layers that are not present in the original sensory signal but arise from the network's own internal dynamics. High-order addictions are defined formally, and their conditions of existence are derived.

Second, we argue that the output of an ANN-Ψ is not, in general, a single tensor event but a *tensorial praxic stream* (TPS): a temporally extended, causally interleaved sequence of partial praxis tensors, each followed by environmental feedback before the next is emitted. We formalize the TPS as a stochastic process, derive the closed-loop dynamics it induces over the environment's state space, and show that classical single-shot output is a degenerate limiting case of the TPS. The combination of hierarchical stimulus generalization and streaming praxic output defines a new computational paradigm we term *Deep Agentic Flow* (DAF).

Contents

1 Introduction

A single Artificial Junky Neuron (AJN) interacts with the world through the simplest possible sensorimotor loop: it receives a stimulus tensor $\boldsymbol{\mathcal{S}}_t$, evaluates its intensity α_t, emits a praxis $\boldsymbol{\mathcal{P}}_t$ via transducers, and observes the perturbed stimulus $\boldsymbol{\mathcal{S}}_{t+1}$. The loop is tight, local, and concrete.

When AJN units are organized into layers and layers are stacked into an ANN-Ψ [2], two qualitatively new phenomena arise that deserve dedicated analysis.

Stimulus propagation and abstraction. The stimulus entering layer $\ell+1$ is not the raw environmental tensor $\boldsymbol{\mathcal{S}}_t^{(0)}$ but a *derived* tensor—the aggregate praxic output of layer ℓ, processed by an intermediate transducer. This derived tensor carries information about layer ℓ's motivational state, not merely the raw sensory signal. As this process repeats across L layers, the stimulus undergoes a succession of transformations that progressively abstract, compress, and recombine the original sensory content. Deep layers "see" a stimulus that is the emergent product of all preceding layers' agentic activities. We show that this process creates conditions for *high-order addictions*: craving attractors that are constituted not by any single feature of the external world but by relational patterns across the network's own internal representations.

Streaming praxic output. A classical neural network maps an input to an output in a single forward pass. An ANN-Ψ operating in an embodied environment cannot afford this luxury: the environment changes continuously, transducers take time to operate, and the feedback from early praxes can reshape the conditions for later ones. The appropriate output model is not a single tensor but a *stream*: a temporally ordered sequence of partial praxis tensors emitted at successive time steps, with environmental feedback arriving between emissions. We formalize this as the Tensorial Praxic Stream (TPS) and derive its closed-loop properties.

The remainder of the paper is organized as follows. Section 2 develops the theory of stimulus tensor propagation and generalization. Section 3 defines high-order addictions and establishes conditions for their emergence. Section 4 characterizes the intricate feedback topology of a running ANN-Ψ. Section 5 introduces the Tensorial Praxic Stream. Section 6 synthesizes both contributions into the Deep Agentic Flow framework and discusses implications.

2 Stimulus Tensor Propagation Across Layers

2.1 Notation and Layer Indexing

We denote by $\boldsymbol{\mathcal{S}}_t^{(\ell)} \in \mathbb{R}^{d_1^{(\ell)} \times \cdots \times d_{k_\ell}^{(\ell)}}$ the stimulus tensor at layer ℓ and time step t, where $\ell = 0$ is the raw environmental input. Layer ℓ contains N_ℓ AJN units, each with its own intensity function $I^{(j,\ell)}$, craving $\mathcal{M}^{(j,\ell)}(t)$, and praxis $\boldsymbol{\mathcal{P}}_t^{(j,\ell)}$.

2.2 The Inter-Layer Stimulus Construction Operator

Definition 1 (Inter-Layer Stimulus Construction)**.** *The* Inter-Layer Stimulus Construction *(ILSC) operator* $\Phi^{(\ell\to\ell+1)}$ *maps the aggregate praxic output of layer* ℓ *to the stimulus* *tensor of layer $\ell + 1$:*

$$\boldsymbol{\mathcal{S}}_t^{(\ell+1)} = \Phi^{(\ell\to\ell+1)}\left(\left\{\boldsymbol{\mathcal{P}}_t^{(j,\ell)}\right\}_{j=1}^{N_\ell},\ \boldsymbol{\mathcal{S}}_t^{(\ell)},\ \boldsymbol{\mathcal{S}}_{t-1}^{(\ell+1)}\right), \tag{1}$$

where the third argument makes the operator memory-augmented*: the newly constructed stimulus depends also on the previous stimulus at that layer, enabling the propagation of temporal context.*

A concrete instantiation of $\Phi^{(\ell\to\ell+1)}$ is the following three-stage composition:

$$\text{(i)}\quad \overline{\boldsymbol{\mathcal{P}}}_t^{(\ell)} = \frac{1}{N_\ell}\sum_{j=1}^{N_\ell} w_j^{(\ell)}\,\boldsymbol{\mathcal{P}}_t^{(j,\ell)}, \tag{2}$$

$$\text{(ii)}\quad \boldsymbol{\mathcal{F}}_t^{(\ell)} = \mathcal{T}^{(\ell)}\left(\overline{\boldsymbol{\mathcal{P}}}_t^{(\ell)}\right), \tag{3}$$

$$\text{(iii)}\quad \boldsymbol{\mathcal{S}}_t^{(\ell+1)} = (1-\rho)\,\boldsymbol{\mathcal{F}}_t^{(\ell)} + \rho\,\boldsymbol{\mathcal{S}}_{t-1}^{(\ell+1)}, \tag{4}$$

where $w_j^{(\ell)} = \mathrm{softmax}(\mathcal{M}^{(j,\ell)})$ are addiction-weighted pooling coefficients (more addicted units contribute more to the layer output), $\mathcal{T}^{(\ell)}$ is the layer transducer, and $\rho \in [0,1]$ is a temporal smoothing coefficient encoding inertia of the inter-layer medium.

Remark 1. *The addiction-weighted pooling in Eq. (2) creates a self-amplifying loop: units with higher craving contribute more to the next-layer stimulus, which in turn shapes the environment toward their preferred class, further reinforcing their craving. This is the first seed of higher-order addiction.*

2.3 Stimulus Dimensionality Transformation

The ILSC operator need not preserve the tensor shape. We distinguish three transformational regimes:

1. **Compression** ($\dim \boldsymbol{\mathcal{S}}^{(\ell+1)} < \dim \boldsymbol{\mathcal{S}}^{(\ell)}$): The transducer $\mathcal{T}^{(\ell)}$ performs a dimensionality reduction, abstracting the praxic output into a lower-dimensional representation. This occurs naturally when $N_{\ell+1} \ll N_\ell$, forcing the upper layer to operate on compressed summaries of lower-layer motivation.

2. **Expansion** ($\dim \boldsymbol{\mathcal{S}}^{(\ell+1)} > \dim \boldsymbol{\mathcal{S}}^{(\ell)}$): The transducer "unpacks" the aggregate praxis into a richer stimulus space by, for instance, applying a learned decoder. This allows upper layers to attend to finer distinctions within the lower layer's motivational output.

3. **Relational fusion** ($\dim \boldsymbol{\mathcal{S}}^{(\ell+1)}$ heterogeneous): When layer ℓ is heterogeneous (multiple stimulus classes), the ILSC operator can concatenate the per-class aggregate praxes along a new class-axis, yielding a stimulus tensor that explicitly encodes inter-class relationships:

$$\boldsymbol{\mathcal{S}}_t^{(\ell+1)} = \left[\overline{\boldsymbol{\mathcal{P}}}_t^{(\ell,1)} \,\|\, \overline{\boldsymbol{\mathcal{P}}}_t^{(\ell,2)} \,\|\, \cdots \,\|\, \overline{\boldsymbol{\mathcal{P}}}_t^{(\ell,K)}\right], \tag{5}$$

where $\|$ denotes concatenation along the new axis. The resulting stimulus tensor at layer $\ell+1$ encodes not just "what class is dominant" but "what is the joint configuration of all classes," enabling upper layers to develop addictions to *multi-class relational patterns.*

3 Emergent High-Order Addictions

3.1 The Stimulus Hierarchy

The successive application of ILSC operators defines a *stimulus hierarchy*:

$$\underbrace{\boldsymbol{\mathcal{S}}_t^{(0)}}_{\text{raw environment}} \xrightarrow{\Phi^{(0\to 1)}} \underbrace{\boldsymbol{\mathcal{S}}_t^{(1)}}_{\text{layer-1 product}} \xrightarrow{\Phi^{(1\to 2)}} \cdots \xrightarrow{\Phi^{(L-1\to L)}} \underbrace{\boldsymbol{\mathcal{S}}_t^{(L)}}_{\text{deep stimulus}} .$$

Each level of the hierarchy is a function of all lower levels' motivational states. We denote by $\mathcal{H}_\ell$ the *stimulus heredity set*: the set of all lower-level stimuli from which $\boldsymbol{\mathcal{S}}_t^{(\ell)}$ is ultimately derived.

3.2 Definition of High-Order Addiction

Definition 2 (Order-ℓ Addiction)**.** *An AJN unit j at layer $\ell > 1$ is said to develop an* order-ℓ addiction *if its craving $\mathcal{M}^{(j,\ell)}(t)$ converges to a stable attractor $\mathcal{M}^* > 0$ in response to a stimulus $\boldsymbol{\mathcal{S}}_t^{(\ell)}$ that has no direct correlate in the raw input $\boldsymbol{\mathcal{S}}_t^{(0)}$—that is, if there exists no function $g : \mathbb{R}^{d^{(0)}} \to [0,1]$ such that $I^{(j,\ell)}(\boldsymbol{\mathcal{S}}_t^{(\ell)}) \approx g(\boldsymbol{\mathcal{S}}_t^{(0)})$ for all t.*

In other words, a high-order addiction is a craving for a pattern that is *constituted by* the network's own transformations of the world, not directly readable from the raw sensory signal. Such addictions are *emergent*: they arise from the network topology, not from the designer's specification.

3.3 Conditions for High-Order Addiction Emergence

Proposition 1 (Sufficient Conditions for Order-ℓ Addiction)**.** *A network ANN-Ψ of depth L will support order-ℓ addictions if and only if all three of the following hold:*

(C1) ***Non-linear transduction.*** *The transducer $\mathcal{T}^{(\ell-1)}$ is not affinely equivalent to $\mathcal{T}^{(0)} \circ \cdots \circ \mathcal{T}^{(\ell-2)}$. Non-linearity breaks the chain of affine equivalences that would otherwise allow the deep stimulus to be written as a linear function of the raw input.*

(C2) ***Temporal memory.*** *The smoothing coefficient $\rho > 0$ in Eq. (4), so that $\boldsymbol{\mathcal{S}}_t^{(\ell)}$ depends on previous time steps. Without memory, each stimulus is a pure function of the current environment, and no genuinely new attractor can form.*

(C3) ***Praxic feedback richness.*** *The transducer bandwidth $\dim(\boldsymbol{\mathcal{F}}_t^{(\ell-1)}) \geq \dim(\boldsymbol{\mathcal{S}}_t^{(0)})$, ensuring that the derived stimulus is at least as informationally rich as the raw input and can therefore encode patterns absent from it.*

Remark 2. *Condition (C1) is satisfied by any non-polynomial transducer (e.g., sigmoidal actuators, physical motors with saturation). Condition (C2) is satisfied whenever $\rho > 0$, even a small inertia suffices. Condition (C3) is a design constraint that requires attention in shallow architectures.*

3.4 The Addiction Emergence Ladder

Under the conditions of Proposition 1, high-order addictions form a *ladder* of increasing abstraction. We characterize each rung:

- **Order 1 (sensory addiction).** The AJN is addicted to a directly observable feature of the environment: a color, a frequency, a spatial gradient. The intensity function $I^{(j,1)}$ is essentially a feature detector on raw input.
- **Order 2 (relational addiction).** The AJN at layer 2 receives a stimulus built from the aggregate praxes of layer-1 units. If those units are heterogeneous (encoding different sensory classes), the order-2 stimulus encodes *co-occurrence patterns* among sensory classes. The order-2 AJN develops a craving for specific relational configurations—not for any single feature.
- **Order 3 (motivational addiction).** At layer 3, the stimulus reflects the *motivational dynamics* of layer 2: which classes are currently dominant, which are in frustration, which in withdrawal. An order-3 AJN is addicted to a particular *pattern of motivation* in the layer below—for instance, to the state in which two specific layer-2 classes are simultaneously in saturation. This is qualitatively closer to a "meta-goal" than a perceptual preference.
- **Order ℓ (abstract addiction).** In general, an order-ℓ addiction is defined over a stimulus space that encodes $\ell - 1$ levels of motivational transformation of the raw environment. These high-order cravings have no direct sensory interpretation but are functionally analogous to high-level cognitive goals: consistency, harmony, novelty, or predictability across the network's own internal states.

Figure 1 illustrates the addiction emergence ladder and the stimulus flow that sustains it.

4 Intricate Feedback Flows in ANN-Ψ

4.1 Three Classes of Feedback

A running ANN-Ψ exhibits not one but three structurally distinct classes of feedback loop, which we now distinguish formally.

Class F1: Immediate Sensorimotor Loop. This is the basic AJN loop defined in Paper I: a unit at any layer emits praxes $\boldsymbol{\mathcal{P}}_t^{(j,\ell)}$, the transducers act on the environment, and the environment returns a modified stimulus. The loop closes at time $t+1$:

$$\underbrace{\boldsymbol{\mathcal{S}}_t^{(\ell)}}_{\text{stimulus}} \to \underbrace{\boldsymbol{\mathcal{P}}_t^{(j,\ell)}}_{\text{praxis}} \to \underbrace{\mathcal{T}^{(\ell)}(\boldsymbol{\mathcal{P}}_t^{(j,\ell)})}_{\text{transducer action}} \to \underbrace{\boldsymbol{\mathcal{S}}_{t+1}^{(\ell)}}_{\text{modified stimulus}} . \tag{6}$$

The delay is one time step; the scope is local to the layer.

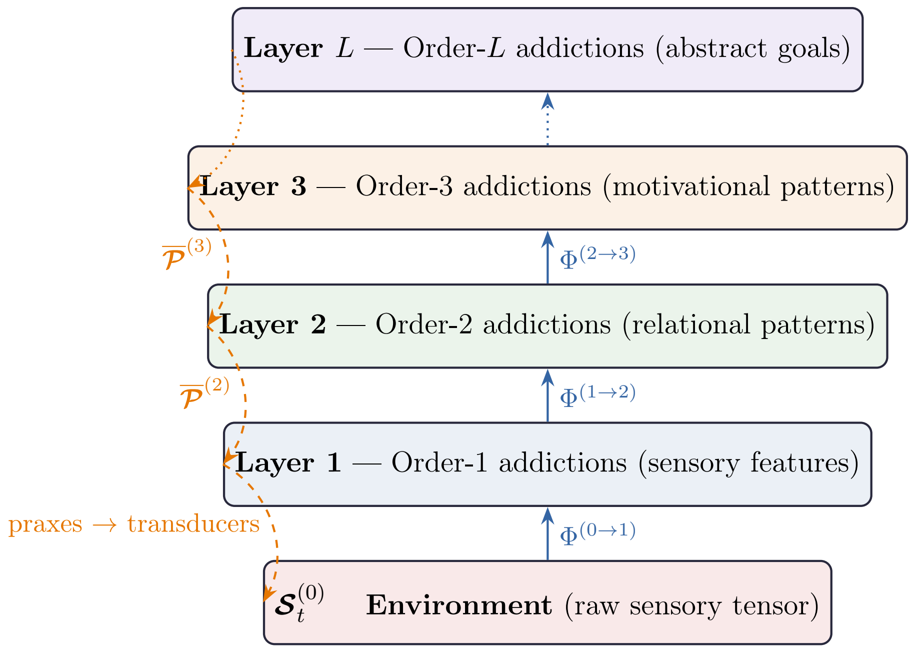

Figure 1: The Addiction Emergence Ladder. Stimuli (solid blue arrows) propagate upward through ILSC operators, increasing in abstraction. Praxes (dashed orange arrows) propagate downward, modulating the stimulus construction at each level. Each layer develops addictions of a higher order than the layer below.

Class F2: Inter-Layer Modulatory Loop. The aggregate praxis of layer ℓ enters the ILSC operator and contributes to the stimulus of layer $\ell+1$. But the craving levels of layer $\ell+1$ units determine the addiction-weighted pooling in Eq. (2) back at layer ℓ, completing a modulatory loop of scope ±1 layer:

$$\mathcal{M}^{(\ell+1)}(t) \xrightarrow{\text{weights } w_j^{(\ell)}} \overline{\mathcal{P}}_t^{(\ell)} \xrightarrow{\Phi} \mathcal{S}_t^{(\ell+1)} \to \mathcal{M}^{(\ell+1)}(t+1). \tag{7}$$

This loop operates at the inter-layer timescale and creates *attentional modulation*: deep-layer addiction states reshape what lower layers collectively emphasize.

Class F3: Long-Range Resonance Loop. When the ANN-Ψ is embedded in a physical environment, the deep-layer praxes (which drive high-order transducers: locomotion, speech synthesis, complex manipulation) modify the raw environmental tensor $\mathcal{S}_t^{(0)}$. This modification propagates upward through all ILSC operators and eventually returns to the deep layer that originally generated the praxis. The loop closes across L layers and an arbitrary number of environmental time steps:

$$\mathcal{P}_t^{(\ell,\text{deep})} \to \Delta\mathcal{S}_{t+\Delta}^{(0)} \xrightarrow{\Phi^{(0\to1)}\circ\cdots\circ\Phi^{(L-1\to L)}} \Delta\mathcal{S}_{t+\Delta}^{(L)} \to \mathcal{M}^{(L)}(t+\Delta+1). \tag{8}$$

The delay $\Delta \gg 1$ creates *temporal credit assignment across the full depth of the network*—the deep unit acts and waits many steps for the consequence to reach it back through the hierarchy. This is the network counterpart of long-horizon planning in reinforcement learning.

4.2 Coupled Resonance Dynamics

The three feedback classes coexist and interact. Let $\mathbf{m}_t = [\mathcal{M}^{(j,\ell)}(t)]$ denote the full craving state vector of all units. The joint dynamics obey a coupled map:

$$\mathbf{m}_{t+1} = \mathbf{G}\left(\mathbf{m}_t,\ \left\{\boldsymbol{\mathcal{S}}_t^{(\ell)}\right\}_{\ell=0}^{L},\ \left\{\boldsymbol{\mathcal{P}}_t^{(j,\ell)}\right\}\right), \tag{9}$$

where $\mathbf{G}$ is the global update operator composed of all F1, F2, F3 sub-functions. The fixed points of $\mathbf{G}$ (when they exist) correspond to *stable high-order addiction configurations*: states in which every layer's craving is either saturated or in a steady withdrawal–reactivation cycle, and the entire network is locked into a self-consistent goal-seeking attractor.

Proposition 2 (Existence of High-Order Attractors). *If the ILSC operators* $\Phi^{(\ell\to\ell+1)}$ *are Lipschitz-continuous with constant* Λ_ℓ*, and if the product* $\prod_{\ell=0}^{L-1}\Lambda_\ell < 1$ *(contraction condition), then the coupled map Eq. (9) admits at least one fixed point* $\mathbf{m}^* \in [0,1]^{\sum_\ell N_\ell}$*, which is a stable high-order addiction configuration.*

The contraction condition is satisfied in practice when the transducers implement saturating non-linearities (sigmoidal actuators, normalized motor commands), which is the natural design for safe physical systems.

5 The Tensorial Praxic Stream

5.1 Motivation: Why a Single Output Tensor Is Insufficient

Classical neural networks treat output as an event: given input x, produce $\hat{y}$ in a single forward pass. This model is adequate for static pattern recognition but fails when:

1. The environment changes faster than the inference time of the full network;
2. The action space requires multiple sequential operations (e.g., a robotic arm must grasp before it can lift);
3. Feedback from early actions is necessary to compute later actions correctly;
4. The "output" is inherently temporal (speech, music, a sequence of movements).

An ANN-Ψ operating in a real environment faces all four challenges simultaneously. The appropriate model is the *Tensorial Praxic Stream*.

5.2 Formal Definition

Definition 3 (Tensorial Praxic Stream). *A* Tensorial Praxic Stream *(TPS) generated by ANN-*Ψ *over a time horizon* $[t_0, t_0+T]$ *is a stochastic process*

$$\Pi = \left\{\left(\boldsymbol{\mathcal{P}}_{t_k},\ \boldsymbol{\mathcal{F}}_{t_k},\ \boldsymbol{\mathcal{S}}_{t_k}^{(0)}\right)\right\}_{k=0}^{K},$$

where:

- $\boldsymbol{\mathcal{P}}_{t_k} \in \mathbb{R}^{m_k}$ *is the* partial praxis tensor *emitted at time step* t_k*;*

- $\mathcal{F}_{t_k} = \mathcal{T}(\boldsymbol{\mathcal{P}}_{t_k})$ *is the transducer output (physical action) at* t_k*;*

- $\boldsymbol{\mathcal{S}}^{(0)}_{t_k}$ *is the environmental state* after *the transducer action has taken effect, i.e., the feedback;*

- *the emission times* $t_0 < t_1 < \cdots < t_K$ *are adapted to the process (they may be random);*

- *each* $\boldsymbol{\mathcal{P}}_{t_{k+1}}$ *is a measurable function of* $\boldsymbol{\mathcal{S}}^{(0)}_{t_0}, \ldots, \boldsymbol{\mathcal{S}}^{(0)}_{t_k}$ *(the causal structure).*

The key structural property of the TPS is its *causal interleaving*: every partial praxis depends on all preceding feedbacks. This distinguishes the TPS from a simple sequence of independent actions.

5.3 TPS Dynamics: The Closed-Loop Equation

The TPS is generated by the following closed-loop recursion, which replaces the single forward pass of classical networks:

$$\boxed{\boldsymbol{\mathcal{P}}_{t_{k+1}} = \Omega_{\text{net}}\Big(\mathcal{H}_{t_k},\ \boldsymbol{\mathcal{S}}^{(0)}_{t_k},\ \mathbf{m}_{t_k}\Big)} \tag{10}$$

where Ω_{net} is the network-level policy (the composition of all layer policies), $\mathcal{H}_{t_k}$ is the full history of stimuli and praxes up to t_k, and $\mathbf{m}_{t_k}$ is the current craving state vector.

The environmental dynamics close the loop:

$$\boldsymbol{\mathcal{S}}^{(0)}_{t_{k+1}} = \mathcal{E}\Big(\mathcal{T}(\boldsymbol{\mathcal{P}}_{t_{k+1}}),\ \boldsymbol{\mathcal{S}}^{(0)}_{t_k}\Big), \tag{11}$$

where $\mathcal{E}$ is the environment transition operator (deterministic or stochastic). Together, Eqs. (10) and (11) define a *stochastic recurrent dynamical system* over the joint state space $(\boldsymbol{\mathcal{S}}^{(0)}, \mathbf{m})$.

5.4 TPS Dimensionality and Emission Rate

An important design parameter is the *emission schedule*: at what intervals $\Delta t_k = t_{k+1} - t_k$ does the network emit partial praxes? Three canonical regimes exist:

1. **Fixed-rate streaming** ($\Delta t_k = \Delta t$ constant): the network emits a new praxis tensor at every Δt time steps, regardless of the feedback content. This is appropriate for real-time motor control where regularity is required.

2. **Event-triggered streaming**: the network emits a new praxis tensor only when the feedback tensor $\boldsymbol{\mathcal{S}}^{(0)}_{t_k}$ changes by more than a threshold ϵ:

$$t_{k+1} = \min\Big\{t > t_k \ :\ \|\boldsymbol{\mathcal{S}}^{(0)}_t - \boldsymbol{\mathcal{S}}^{(0)}_{t_k}\|_F > \epsilon\Big\}. \tag{12}$$

   This conserves computational resources and is appropriate for sparse, high-impact environmental changes.

3. **Addiction-driven streaming**: the emission schedule is modulated by the craving state. A unit approaching saturation ($\alpha_t \to \alpha_{\max}$) *suppresses* its emission (Phase 3 in the AJN cycle), while a unit in frustration (Phase 5) *accelerates* its emission:

$$\Delta t^{(j)}_{k+1} = \Delta t_{\min} + (\Delta t_{\max} - \Delta t_{\min}) \cdot \frac{\alpha^{(j)}_{t_k}}{\alpha_{\max}}. \tag{13}$$

5.5 The TPS as a Generalization of Classical Output

The classical single-shot output is recovered as a degenerate limit of the TPS:

Proposition 3 (Classical Output as TPS Limit)**.** *Let $K = 1$ (a single emission), $\rho = 0$ (no temporal memory in ILSC), and $\Delta = 0$ (no delay in the long-range loop). Then the TPS $\Pi = \{(\boldsymbol{\mathcal{P}}_{t_0}, \boldsymbol{\mathcal{F}}_{t_0}, \boldsymbol{\mathcal{S}}^{(0)}_{t_1})\}$ reduces to the single forward pass of a classical neural network: $\hat{y} = f(x) \equiv \boldsymbol{\mathcal{F}}_{t_0}$ where $x \equiv \boldsymbol{\mathcal{S}}^{(0)}_{t_0}$.*

This shows that the TPS is a strict generalization: every classical neural network output is a TPS with $K = 1$, but TPS with $K > 1$ and $\rho > 0$ encode qualitatively richer input–output relationships.

5.6 Information Flow in the TPS

Define the *TPS information tensor* at step k as:

$$\mathcal{I}_{t_k} = \bigoplus_{s=0}^{k} \gamma^{k-s} \cdot I\left(\boldsymbol{\mathcal{S}}^{(0)}_{t_s}\right), \tag{14}$$

where $\gamma \in (0, 1]$ is a temporal discount factor and $\oplus$ denotes channel-wise accumulation. The TPS information tensor is a compressed history of all past stimulus intensities, serving as a *temporal context signal* that modulates the generation of future partial praxes. When $\gamma = 1$ no past information is discounted; when $\gamma \to 0$ only the most recent feedback matters.

6 Deep Agentic Flow: Synthesis

6.1 The DAF Framework

We now synthesize the two contributions of this paper into a unified framework.

Definition 4 (Deep Agentic Flow)**.** *A* Deep Agentic Flow *(DAF) is the joint process:*

$$DAF = \left(ANN\text{-}\Psi,\ \left\{\Phi^{(\ell\to\ell+1)}\right\}_{\ell=0}^{L-1},\ \Pi,\ \mathcal{E}\right),$$

consisting of an ANN-Ψ architecture, a stimulus hierarchy defined by ILSC operators, a TPS Π governing the output, and an environment $\mathcal{E}$. DAF operates as a perpetual closed loop: the TPS acts on $\mathcal{E}$, $\mathcal{E}$ returns stimuli that enter the bottom of the stimulus hierarchy, the hierarchy propagates and transforms these stimuli upward, deep layers develop and maintain high-order addictions, and the resulting network craving state drives the next emission of partial praxes.

6.2 The DAF Computation Graph

Figure 2 presents the complete DAF computation graph, showing the interplay of the three feedback classes with the stimulus hierarchy and TPS.

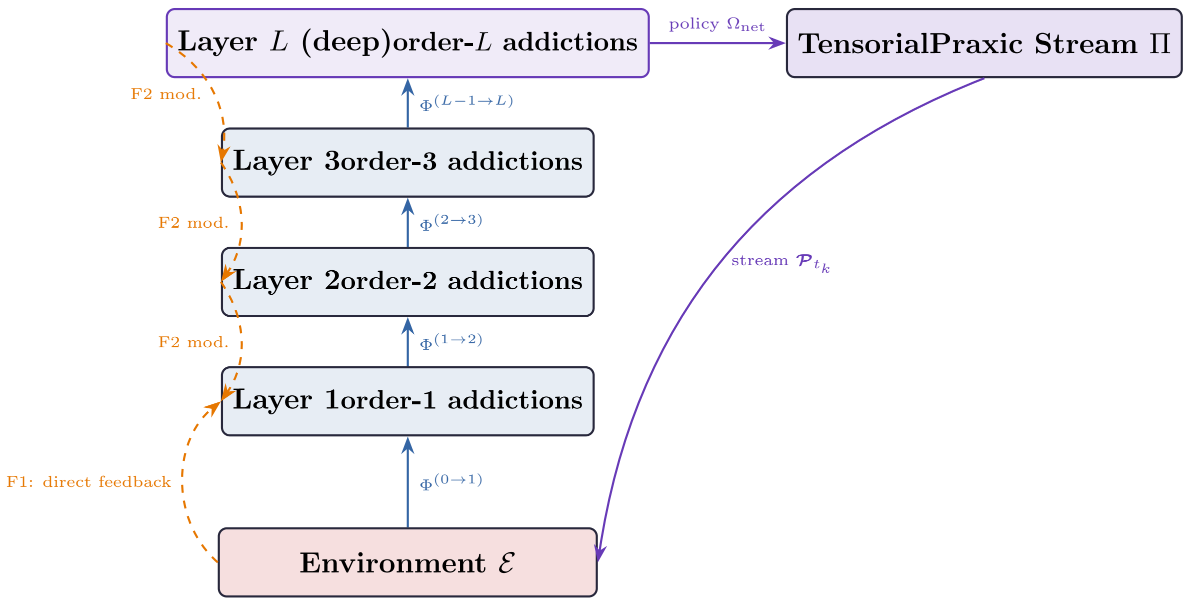

Figure 2: The Deep Agentic Flow computation graph. Solid blue arrows: stimulus propagation via ILSC operators. Dashed orange arrows: feedback flows (F1 direct, F2 modulatory). Solid purple arrows: TPS emission and long-range loop (F3). The deep layer drives the TPS, whose partial praxes act on the environment and return as new stimuli at the bottom of the hierarchy.

6.3 DAF Algorithm

Algorithm 1 Deep Agentic Flow — Full Operational Cycle

Initialize: ANN-Ψ state $(\mathbf{m}_0, \boldsymbol{\mathcal{S}}_0^{(0)}, \{\boldsymbol{\mu}_0^{(j,\ell)}\})$; emission schedule; $k \leftarrow 0$
while not terminated **do**
  for $\ell = 1$ to L **do**
    Compute $w_j^{(\ell-1)} \leftarrow \mathrm{softmax}(\mathcal{M}^{(j,\ell-1)})$
    Pool: $\overline{\boldsymbol{\mathcal{P}}}_{t_k}^{(\ell-1)} \leftarrow \sum_j w_j^{(\ell-1)} \boldsymbol{\mathcal{P}}_{t_k}^{(j,\ell-1)}$ // *Eq.* (2)
    Transduce: $\boldsymbol{\mathcal{F}}_{t_k}^{(\ell-1)} \leftarrow \mathcal{T}^{(\ell-1)}(\overline{\boldsymbol{\mathcal{P}}}_{t_k}^{(\ell-1)})$
    Construct: $\boldsymbol{\mathcal{S}}_{t_k}^{(\ell)} \leftarrow (1-\rho)\boldsymbol{\mathcal{F}}_{t_k}^{(\ell-1)} + \rho\, \boldsymbol{\mathcal{S}}_{t_k-1}^{(\ell)}$ // *Eq.* (4)
  end for
  for $\ell = 1$ to L, $j = 1$ to N_ℓ **do**
    Run AJN cycle (Algorithm 1, Paper I) with input $\boldsymbol{\mathcal{S}}_{t_k}^{(\ell)}$
    Update $\mathcal{M}^{(j,\ell)}$, $\theta^{(j,\ell)}$, $\boldsymbol{\mu}^{(j,\ell)}$, $\Sigma^{(j,\ell)}$
  end for
  Determine emission time t_{k+1} via chosen schedule (Eqs. (12) or (13))
  Compute $\boldsymbol{\mathcal{P}}_{t_{k+1}} \leftarrow \Omega_{\text{net}}(\mathcal{H}_{t_k}, \boldsymbol{\mathcal{S}}_{t_k}^{(0)}, \mathbf{m}_{t_k})$ // *Eq.* (10)
  Execute $\mathcal{T}(\boldsymbol{\mathcal{P}}_{t_{k+1}})$ on $\mathcal{E}$
  Observe $\boldsymbol{\mathcal{S}}_{t_{k+1}}^{(0)} \leftarrow \mathcal{E}(\mathcal{T}(\boldsymbol{\mathcal{P}}_{t_{k+1}}), \boldsymbol{\mathcal{S}}_{t_k}^{(0)})$ // *Eq.* (11)
  Propagate $\boldsymbol{\mathcal{S}}_{t_{k+1}}^{(0)}$ upward via ILSC operators (will take effect in next iteration)
  $k \leftarrow k+1$
end while

6.4 Emergent Temporal Structure of the TPS

A DAF running under addiction-driven streaming (Eq. (13)) exhibits a characteristic temporal structure that reflects the collective addiction state of the network:

- **Burst phases**: when the deep layer is in frustration (Phase 5), $\Delta t_{k+1} \to \Delta t_{\min}$, and the TPS emits a rapid burst of partial praxes—high-frequency, high-magnitude actions exploring the environment. This corresponds to what an observer would describe as *agitated, searching behavior*.
- **Quiescent phases**: when the deep layer is in saturation (Phase 3), $\Delta t_{k+1} \to \Delta t_{\max}$, and the TPS emission slows dramatically. The network acts minimally, maintaining its achieved state with low-energy maintenance praxes. This corresponds to *resting, satisfied behavior*.
- **Rhythmic phases**: when the deep layer oscillates between withdrawal and reinforcement (Phases 4 and 2), the emission rate oscillates accordingly, producing a *rhythmic, exploratory pattern*—periodic probing of the environment with moderate-energy praxes.

These three behavioral modes emerge purely from the AJN addiction dynamics; no external controller or scheduler is required.

7 Discussion

7.1 High-Order Addictions and Cognitive Analogy

The addiction emergence ladder (Section 3) has a provocative interpretation when extended to many layers. An order-1 addiction is analogous to a perceptual preference; an order-2 to a relational concept ("I crave the co-occurrence of A and B"); an order-3 to a social or contextual expectation ("I crave the state in which my subsystems are all satisfied simultaneously"). At very deep layers, the addiction object becomes what one might loosely call a *value*: a stable internal attractor defined entirely in terms of the network's own representational hierarchy.

This suggests that the Agentic Theory provides a *bottom-up* account of the emergence of abstract goals and values from simple low-level craving dynamics, without requiring any top-down specification of what the system "should" want.

7.2 TPS and Embodied Cognition

The TPS formalization aligns with the enactivist and embodied cognition traditions [7], which argue that cognition is not input–output mapping but a continuous, environmentally coupled process. The TPS makes this precise: the network never "finishes" computing its output; it continuously emits partial actions and absorbs partial feedbacks, with no clear boundary between perception and action.

7.3 Relation to Continuous Control and Recurrent Architectures

The DAF framework can be seen as a principled extension of continuous-action reinforcement learning and recurrent neural networks. Compared to LSTM- or GRU-based architectures [5], the DAF temporal memory is not stored in gated cell states but distributed across the craving levels of all AJN units—a fundamentally different memory substrate. Compared to policy gradient methods [8], the praxic policy update operates locally at each unit and does not require a global advantage estimate.

7.4 Open Problems

1. **ILSC learning.** The ILSC operators $\Phi^{(\ell\to\ell+1)}$ are currently fixed at design time. Learning them end-to-end—discovering what transformations best support high-order addiction formation—is an open and important problem.

2. **TPS truncation and horizon selection.** The DAF runs indefinitely; real systems require principled criteria for truncating the TPS ("when is the task done?"). Defining a network-level stopping condition in terms of collective saturation is a natural direction.

3. **Multi-agent DAF.** If multiple ANN-Ψ systems share the same environment, their TPSs interact: one network's praxes modify the stimuli available to the others. The emergent dynamics of multi-agent DAF—coalition, competition, communication—constitute a rich and unexplored domain.

4. **Stability and controllability of high-order attractors.** While Proposition 2 guarantees the existence of attractors under mild conditions, their stability margins, bifurcation structure, and controllability (can we steer the network from one high-order addiction configuration to another?) remain open questions.

8 Conclusion

This paper has developed two foundational extensions to the Agentic Theory.

First, we showed that a stimulus tensor entering an AJN layer is not simply consumed; it is *transformed*, *compressed or expanded*, and *passed upward* through a hierarchy of ILSC operators. Each level of the hierarchy constructs a richer, more abstract stimulus space, enabling the emergence of high-order addictions—stable craving attractors defined over patterns that are purely internal products of the network's own dynamics and have no direct sensory correlate. The addiction emergence ladder formalizes this process and connects it to the three classes of feedback loop that characterize a running ANN-Ψ.

Second, we argued that the output of an ANN-Ψ is not a single tensor event but a Tensorial Praxic Stream: a causally interleaved sequence of partial praxis tensors, each conditioned on the environmental feedback from all preceding ones. We defined the TPS formally, derived its closed-loop dynamics, characterized its three canonical emission regimes, and showed that classical single-shot output is a degenerate limit of the TPS.

Together, these two contributions define Deep Agentic Flow: a framework in which the network's internal motivational hierarchy and its temporal behavioral stream are co-constituted through the perpetual closed loop of action, feedback, and recursive stimulus transformation. We believe DAF provides a principled new language for describing adaptive, embodied, goal-directed artificial systems whose goals are not programmed but *grown*.

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